

From Soil to Carbon

A Spatial Decision-Support Tool for Estate Afforestation

Where to plant, which species, and what the carbon is worth — validated against the UK Woodland Carbon Code.

A. THE CHALLENGE

West Dean Estate spans 3,371.8 hectares in the South Downs National Park and has committed to woodland creation for net zero. Yet its only certified carbon scheme covers just 30.69 ha. No existing tool tells the estate where it can plant, which species suit each soil, and what the carbon is worth — together, at the parcel level.

F. A TOOL BUILT TO ARGUE WITH ITSELF

It doesn't prescribe decisions; it reveals conflicting good options and makes the estate justify its choice. The trade-off becomes impossible to ignore.

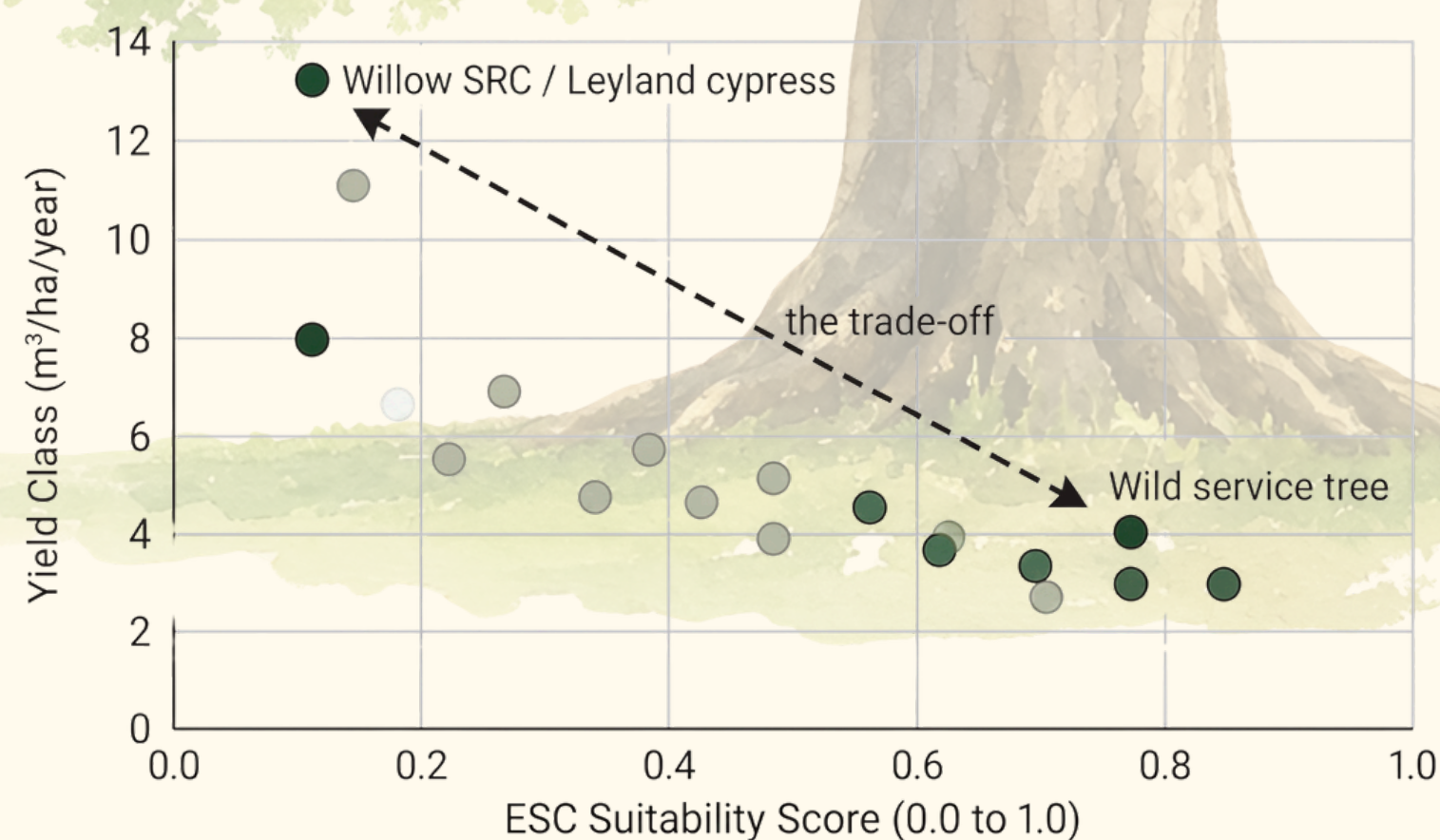
Scenario A →
more carbon

West Dean
decides

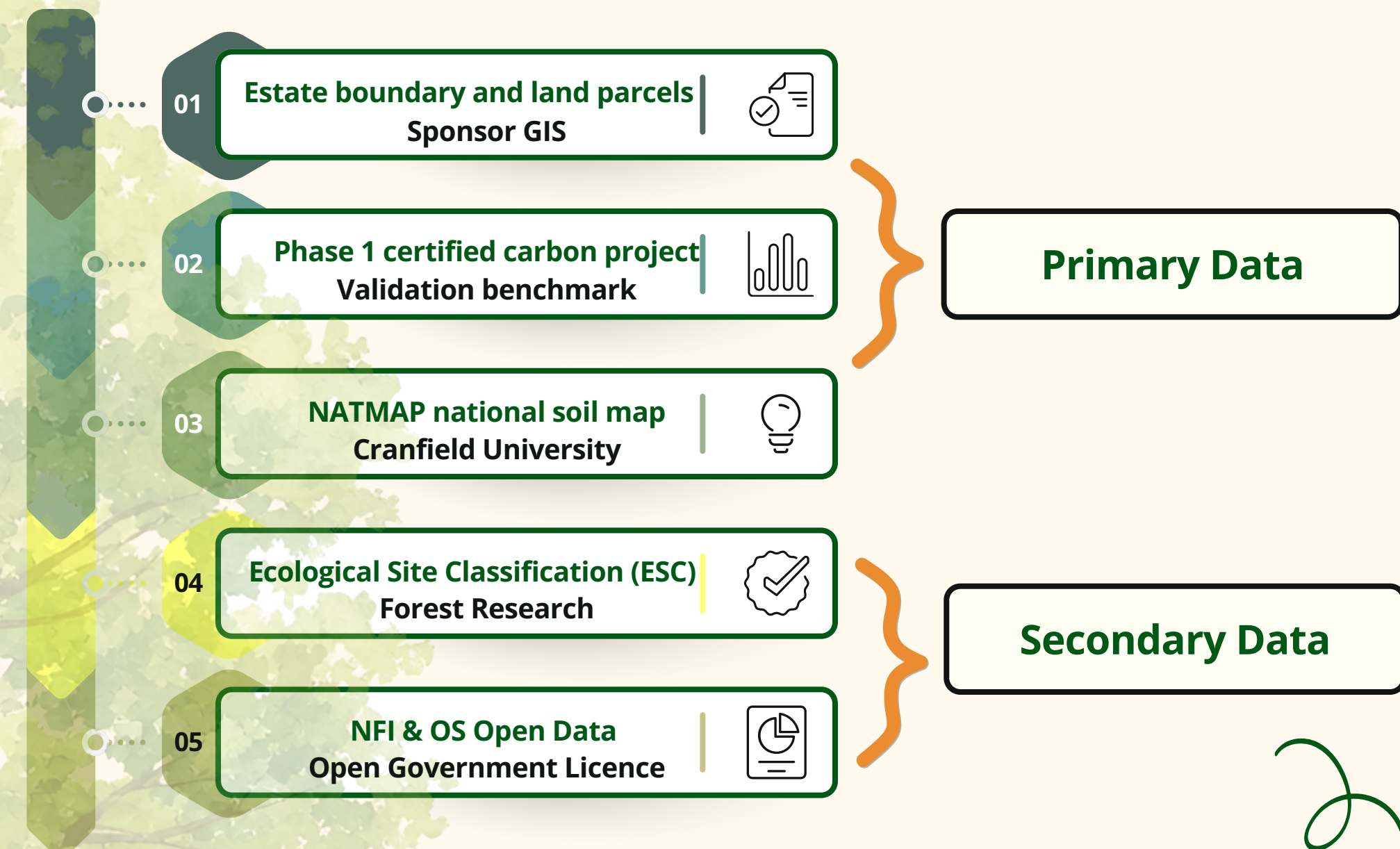
Scenario B →
more suitability

E. THE 18% QUESTION

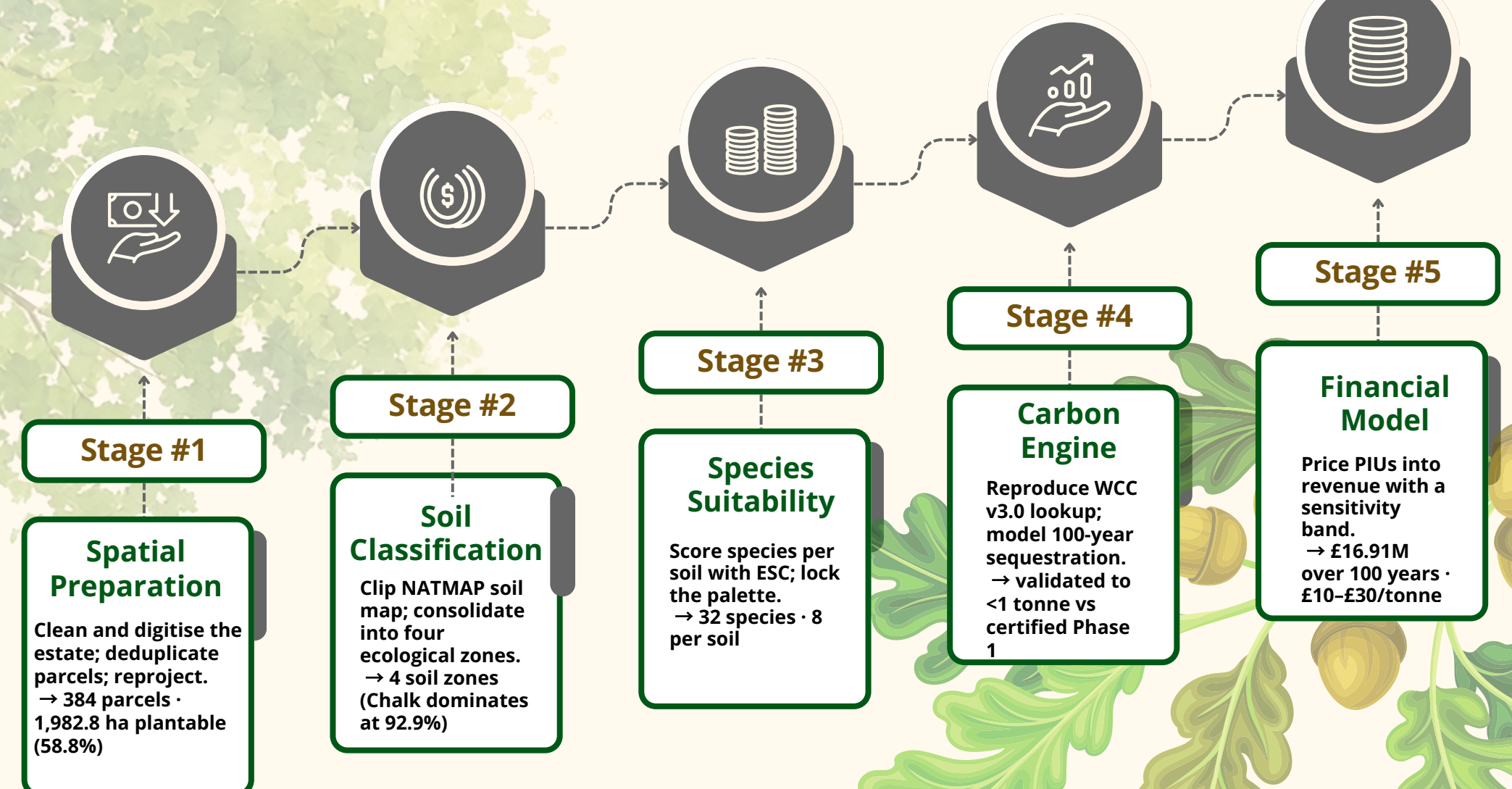
Plant the species the ecologist ranks highest, and you sequester 18% less carbon per hectare. That's not a modelling fault, it's how the Woodland carbon Code is built.



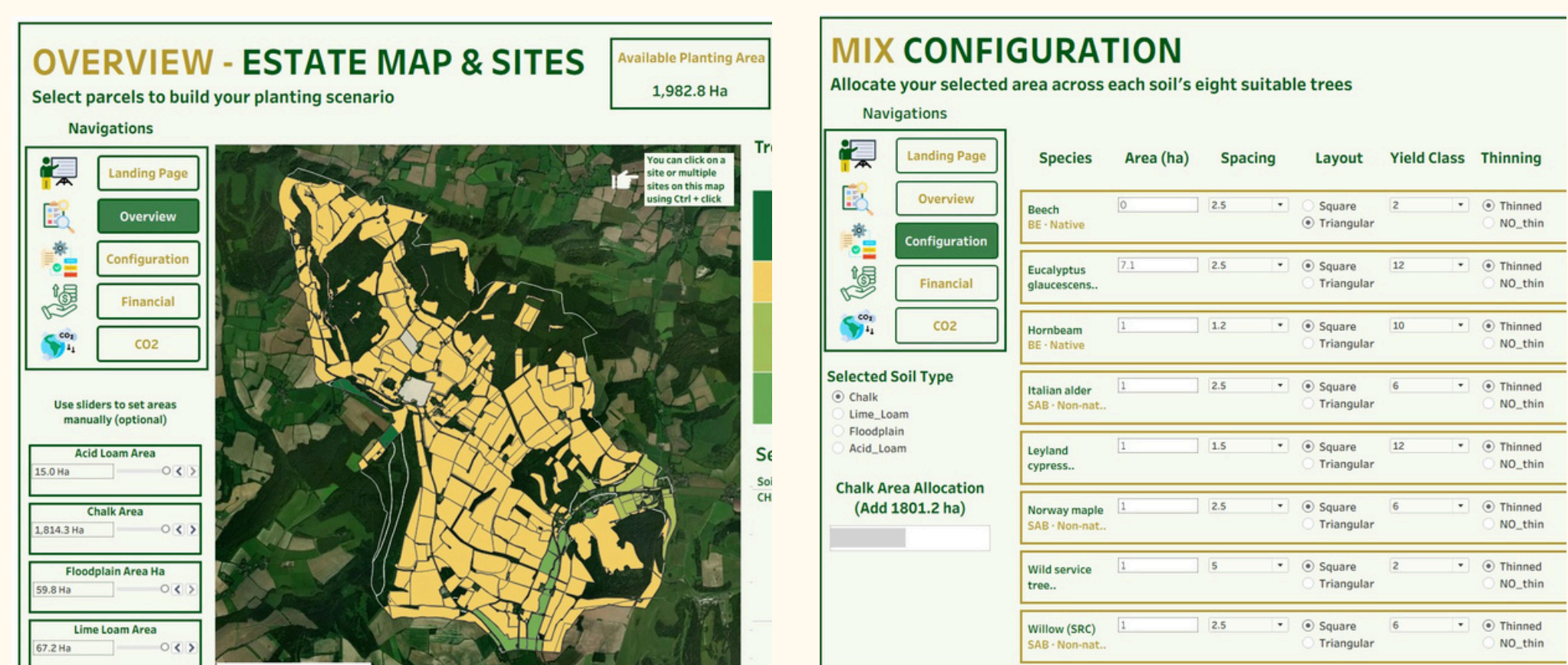
B. RESEARCH DESIGN & DATA



C. DATA TO MODEL



D. FROM BOUNDARY TO DECISION, IN FOUR SCREENS



Project Timeline

Phase 1
Week 1 – Week 3

Data Preparation & Methodology

Cleaned and deduplicated the estate parcels. Classified soils into four zones and locked a 32-species palette using ESC scores.

Phase 3
Week 8 – Week 10

Tool, Results & Write-Up

Built the five-dashboard Tableau tool, ran both scenarios, and drew out the carbon-versus-ecology trade-off. Finalised the report and poster.

Foundation & Literature

Scoped the three research questions and reviewed literature on carbon accounting, soil variation, and species matching. Gathered the estate boundary, soil, and species data.

Phase 2
Week 4 – Week 7

Model Build & Validation

Built the WCC carbon-quantification engine and PIU revenue chain. Validated it against West Dean's certified Phase 1 project to within one tonne.

Phase 4
Week 11 - Week 14

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